The Medium

A Foundational Theory of Reality

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2026

1 Introduction

1.1 The Core Problem

Physics inherited a set of primitives. Object, location, motion, sequential time, the detached external observer. These were abstracted from human-scale perceptual experience the cave dweller's sky, formalized. Every subsequent framework inherited them without examination. Thermodynamics, electromagnetism, relativity, quantum mechanics, string theory, loop quantum gravity. Each built on the same foundation.

The incompatibilities between frameworks GR against QM, the measurement problem, the problem of time are symptoms of foundational mismatch. Not technical failures awaiting mathematical fixes.

1.2 The Language Problem

Quantum mechanics borrowed ordinary language and bent each term. Observer no longer means a conscious mind. Measurement no longer means careful observation. Collapse may not be a physical event at all. The borrowed terms carry associations the new meanings cannot support. False familiarity misleads and gatekeeps critique.

The Medium requires its own vocabulary. Three terms replace the inherited ones:

- *Occurrence*: a event the unit of substrate process
- *Registration*: retention of a structural consequence of interaction
- *Boundary*: the interface at which two processes alter each other's state

Chosen to mean exactly what they say. Nothing more.

1.3 A Recognition and a Challenge

Modern science is one of the great achievements of the human mind. Quantum mechanics predicts the behavior of matter at the smallest accessible scales with a precision unmatched in the history of empirical inquiry. General relativity describes the geometry of spacetime and the behavior of gravity across cosmological distances with equal exactness. The Standard Model catalogs the fundamental particles and their interactions with extraordinary completeness. These are not approximations or guesses. They are hard-won structures — the accumulated result of centuries of observation, mathematical invention, and experimental testing.

This document does not dispute any of it.

What it asks instead is the question: *What alternative foundation explains the findings of modern science?* The equations of physics are precise maps. The challenge is to identify the territory they map — not the territory of observable phenomena, which physics has surveyed magnificently, but the foundation that gives rise to a universe in which observation is possible at all.

There is a limitation that physics has not confronted fully. Anyone who has ever done physics is an earth-bound observer, even with the advent of space-borne sensors. That observer arrived late — billions of years after the assumed beginning of the universe — in the midst of an apparently infinite universe. That observer, equipped with sensory tools required for survival, began to ask the questions. Using a dimensional coordinate system (meters, seconds, joules), physics constructed systems defined to understand what is seen from that observation point. Constants, dimensions, and equations carry the fingerprints of that position.

> *Out of an infinite nothing all of creation becomes.*
>
> That sentence is the theory. Everything that follows is the attempt to say it precisely.

2 Abstract

The Medium is a foundational framework prior to the limits of human observation. It cannot be tested or demonstrated directly. It is offered as a starting point upon which others may build.

The sole primitive is ι: the Medium's act of self-distinction. The Medium distinguishes within itself. ι is that act. Each ι event produces character — the difference the distinction makes. Character is not a property ι holds. It is what distinguishing is. ι is not point-like. It has multiple relational modes — qualitatively distinct aspects of the ι event itself, not binary opposition. ι has intrinsic cohesive attraction: not because it is complementary to another ι event, but because attraction is what ι is. From intrinsic attraction, self-organization follows without instruction. Without difference there is no distinction. The Medium is the endless occurrence of ι, unconstrained. No boundary, structure, or assumed constraint exists at the substrate level. Space, time, dimensionality, and causality are not conditions imposed upon ι — they emerge from ι-relations. The projection boundary marks the threshold at which the substrate meets the observer's dimensional framework. In the Medium's unit system, one ι event = 1. What SI physics calls $\hbar$ is the SI measurement of that unit — not a fundamental constant, but a unit conversion artifact locating the projection boundary on the human-scale SI grid.

The central organizing claim of v0.5: processes occur independently. No ordering. No coordination required. Yet existence may impact existence. Sequence is what this looks like from inside an embedded process. The substrate field ϕ records distinction characters across the Medium.

From distinction characters, a three-level algebraic hierarchy is derived by successive combination. At the first level, $\kappa = \sum \iota$ is the coupling character of a local collection — the net distinction character of a group of ι events. At the second level, $\mathcal{K} = \sum \kappa$ is the total configuration energy of a candidate stable structure. At the third level, $\mathcal{K}$ projects into spacetime as $\hat{H}$, the Hamiltonian of standard physics — not assumed but derived from substrate structure.

Valid combinations are governed by a logical adjacency constraint: two ι events of identical distinction character produce no new information and therefore cannot occupy adjacent positions in a combination. This is not a physical rule imposed on ι but a logical necessity — identical adjacent distinction characters produce nothing new. The combinatorial space of valid configurations is sharply restricted by this constraint. A minimum threshold of combinatorial complexity must be reached before any stable configuration can form — below this threshold, combinations approach but do not achieve coherence. At the view layer, these sub-threshold combinations appear as quantum vacuum fluctuations. The measurable consequences of the quantum vacuum — the Casimir force between conducting plates, the Lamb shift in hydrogen energy levels — are view-layer QED results. Whether the substrate

fluctuation account produces the correct vacuum pressure and energy perturbation for those measurements is an open derivation carried in Unresolved Questions.

Stable configurations emerge where $\mathcal{K}$ reaches a local extremum. These are the substrate analogs of bound states. Mass, charge, and spin are not independent intrinsic properties — they are three projection signatures of a single underlying combinatorial structure. Mass corresponds to N as a first approximation — larger configurations project greater mass. The specific mass values and the hierarchy between generations are not yet derived. Charge corresponds to the net encounter character of $\mathcal{K}$. Spin describes the relational closure of a configuration's encounter pattern within the web. It is not rotation.

The Medium distinguishes three categories of emergent property. Properties emergent from the substrate arise directly from ι-combinations: mass, charge, spin, the quantum vacuum, the Hamiltonian, spacetime, and dimensionality. Properties emergent at the view layer arise from interactions between stable configurations in spacetime: the fundamental forces, fields, and the full complexity of observable reality. This is the domain of existing physics — The Medium makes no claims there. A third category remains unassigned: color charge, gravity, neutrino oscillation, and dark matter have not yet been placed in either layer with confidence. Dark energy is treated separately — it dissolves as a projection artifact of misread redshift rather than requiring substrate explanation. Gravity is the most consequential of these — if spacetime is itself a projection from ι-relations, gravity may bridge both layers in a way the other forces do not.

The four major puzzles of standard cosmology — the entropy problem, the arrow of time, cosmological redshift, and the singular origin of the Big Bang — are addressed in Part III. In each case the puzzle dissolves rather than being solved, because it was generated by a category error that The Medium does not commit: treating view-layer artifacts as substrate-level facts.

Since ι-arrangements are below the current observational floor, direct testing is not possible with existing instruments. The framework predicts that particle properties treated as intrinsic by standard physics are relational outcomes of substrate combinatorics. New experimental approaches designed to probe transition signatures rather than particle states directly are identified as the path toward indirect verification.

The central open problems are offered as challenges to the theoretical physics community: deriving the specific configurations corresponding to observed particles; establishing which ordered combinations produce which spin values under projection; determining the minimum threshold from first principles; resolving whether color charge and gravity are substrate or projection phenomena; and characterizing dark matter candidates within the substrate. Dark energy is treated as a dissolved projection artifact rather than an open substrate problem.

Changes from v0.4: Reorganized around the independence primitive as the central organizing claim. Revised ι: occurrence without attribute; attribute produced at encounter, not carried by ι. Encounter established as the primitive event from which relational character and particle properties derive. Topos theory connection placed at the encounter level. Added Part IV: What The Medium Needs to Prove. Added new vocabulary layer (occurrence, registration, boundary) to replace inherited QM terminology. Added Observer and Observed section dissolving the observer/observed asymmetry at the substrate level. Expanded time section with the real-for-the-observer vs. necessary-for-the-universe distinction. Added The Core Problem and The Language Problem as Introduction subsections. Revised ι: not point-like; multiple relational modes, not binary opposition; intrinsic cohesive attraction as a property of

ι itself. Mass identified as a stable stochastic imperfection in the ι configuration; gravity as the strain field propagating through the medium toward ground-state restoration. Relational web reframed as a probability landscape — stochastic all the way down, statistical consequence not active process. GR/QM conflict dissolved across the three layers: substrate granularity, relational-web averaging, view-layer smoothness. Wave function and collapse dissolved via the probability landscape account.

3 PART I: THE SUBSTRATE

The substrate description is complete in itself. It contains no dimensions, no observers, no physics. It is what it is.

3.1 The Independence Primitive

Processes occur independently. No ordering. No coordination required. Yet existence may impact existence.

Sequence is what independence looks like from inside an embedded process. Experienced as before and after. The experience is real. The sequence is not the substrate.

Identity is produced at encounter. Each process occurs on its own terms. The relationships are what the processes are. Remove the web and there is no process. Remove the process and there is no web.

Three prior frameworks reach toward this ground and stop short.

String theory places a string at the foundation — a one-dimensional entity carrying vibration modes intrinsically. Each frequency corresponds to a different particle. Spacetime is already there before the string. It is the container. Properties are in the entity, not the encounter.

Loop quantum gravity makes spacetime emergent from a spin network — nodes and edges that are atoms of geometry. That is a genuine step. But the network structure itself is assumed, not derived. LQG does not explain why that relational structure exists. It starts one level too late.

Planck bits — Wheeler's *It from Bit*, Fredkin, Wolfram — place a discrete bit at the foundation carrying 0 or 1 intrinsically. The binary distinction is assumed before any encounter. The question of which value is answered before the question is formed.

ι is none of these. All three prior frameworks place entities first and derive relationships. The Medium places the act of distinction first. Not an entity. Not a web requiring prior connection. The Medium distinguishes within itself. ι is that act. Character is what the act produces.

Planck bits come closest — both are discrete, dimensionless, pre-spatial. The gap is not small. A bit carries its state intrinsically. ι produces character as the act of distinction. That is the difference between assuming distinction and deriving it.

Loop quantum gravity comes closest structurally — both make spacetime emergent from relational structure. But LQG presupposes the relational structure. The Medium derives it from the act of self-distinction.

3.2 The Foundational Claim

The framework rests on a single foundational claim:

> The Medium always is and always was and always will be. It has no beginning and no end. Intrinsic distinction is an eternal property of the Medium — not an event that occurred but a fundamental condition. The structure we observe is not the result of something that happened. It is the expression of the Medium always happening.

This claim does three things at once.

It eliminates the need for an initial starting point. The question of what initiated the universe only arises with the assumption of time. If the Medium always is and always was, there is no before and no starting point. The trigger needed for a beginning dissolves by identifying it as a consequence of a false assumption.

It eliminates time as a primitive. The Medium exists outside of time. What we call time is not a property of the Medium. It is downstream — a consequence of repeated manifestation, not a fundamental feature of reality.

It re-frames differentiation. The emergence of structure is not a process that started. It is a condition that always is. The ground state and the differentiated state coexist within the Medium eternally, because intrinsic distinction — ι — is eternal.

3.3 The Ground State

The Medium at its ground state is a fully autonomous stochastic system. It requires no inputs, no external cause, no temporal framework, and no governing distribution. Its randomness is not randomness within a framework — it is randomness without framework. No privileged outcome. No preferred state.

This is not chaos. Chaos retains structure. The ground state of the Medium precedes structure entirely. It is meta-randomness — randomness that is random about its own nature.

In a medium of infinite extent with no preferred states, every possible configuration has nonzero probability. This is not intuitive but is mathematically unavoidable. The universe is not something that happened against the odds. It is a necessary expression of a medium that has no mechanism for suppressing any possibility.

3.4 Axioms

3.4.1 Axiom I — The Eternal Medium

The Medium always is and always was. It has no beginning, no end, and no external cause.

$$\nexists\, t_0 : M \text{ begins at } t_0 \qquad \nexists\, t_1 : M \text{ ends at } t_1 \tag{1}$$

Status: Axiomatic.

3.4.2 Axiom II — Intrinsic Distinction

The Medium possesses one property: intrinsic distinction, denoted ι. It is not an operator, not a function, not a process. It is what the Medium is. The Medium distinguishes within itself. ι is that act of self-distinction. Each ι event produces character as the difference the distinction makes. Without difference there is no distinction. Character is not a property ι holds — it is what distinguishing is.

Status: Axiomatic.

3.5 Primitives

3.5.1 Primitive I — The Field

ι is the Medium's act of self-distinction. Each ι event produces character — the difference the distinction makes. That character is not a property ι holds before the act. It is what the act is. +1 and -1 are the two aspects of one distinction. It has no spatial extent, no duration, no location, no mass, no charge, no spin. It is dimensionless.

Each ι event occupies a relational node — a position defined entirely by its co-manifestation relations. Not by spatial coordinates. Spatial position is what a relational node looks like under projection.

The substrate field ϕ records the distinction character produced by each ι event. M here is the set of relational nodes, not a spatial manifold.

$$\phi : M \to \{+1, -1\} \tag{2}$$

A record of distinction characters. The complete description of the Medium's state.

At the view layer, one ι event = 1 in the Medium's unit system. The SI physics community locates this threshold at the Planck scale — the scale at which quantum and gravitational effects are simultaneously relevant. That location is an empirical correspondence, not a definition.

ι events are not driven. No prior state determines which relational node the Medium next distinguishes within itself. They simply occur.

Status: Revised in v0.5.

3.5.2 Primitive II — Distinction and Character

Each ι event is an act of self-distinction within the Medium. The character it produces — +1 or -1 — is the difference the distinction makes. Not determined in advance. Not driven by prior state. Whether the Medium next distinguishes at a given relational node is not governed. It simply does.

$$\phi(\iota) \in \{+1, -1\} \text{ (character of distinction, no preference, no guarantee)} \tag{3}$$

The substrate imposes no preference on which character is produced. Without differential distinction character, ι events produce only undifferentiated noise. No structure, no stability, no material. Differential character is the logical necessity for material existence.

A note on the binary axiom. The choice of $\{+1, -1\}$ is defended on logical grounds only: one act of distinction creates one boundary; one boundary has exactly two sides. That argument does not appeal to quantum mechanics or observed discreteness at the Planck scale. Those observations are downstream — consistency checks, not justification. To justify binary ι by reference to quantum discreteness would embed the observable into the foundation the framework is attempting to derive. The axiom stands or falls on the logical claim alone. Whether that claim is sufficient — whether one act of distinction necessarily produces exactly two values rather than three or a continuum — is an open question carried in Unresolved Questions.

This is where topos theory bears on the framework. Topos theory allows logic in which a question has no answer until the context that makes the question meaningful exists. Which character a ι event produces is not determined before the event. The question does not exist until the distinction occurs. This is not epistemic. The answer does not exist yet. That is the logical territory topos theory describes, and it is the logical territory ι occupies.

Shared state. No distance. No space. No ordering.

This is prior to locality. Any spatial relationship between x and y is constructed downstream, after structure formation establishes the conditions in which distance becomes meaningful.

Status: Revised in v0.5.

Not a beginning and an end. A breathing.

Because ι is an eternal property of the Medium, this arc does not occur once. It occurs wherever and whenever ι manifests. Encounter patterns form, achieve stability, dissolve — across an infinite relational web that does not notice.

3.6 The Algebraic Hierarchy

The structure of the Medium generates a natural algebraic hierarchy through successive combination. Each level is derived from the level below it; none is assumed.

Level 0 — The Primitive. ι is the sole primitive. The Medium's act of self-distinction. Each ι event produces character — the difference the distinction makes. +1 and -1 are the two aspects of one distinction event. Dimensionless. Pre-spatial. Pre-temporal.

Level 1 — Coupling Character. For a local collection of ι events, the coupling character κ is defined as their sum:

$$\kappa = \sum_i \iota_i \tag{4}$$

κ is the net distinction character of a local cluster. It is the first derived quantity in the hierarchy.

Level 2 — Configuration Energy. For a candidate stable structure composed of multiple κ clusters, the total configuration energy $\mathcal{K}$ is:

$$\mathcal{K} = \sum_j \kappa_j \tag{5}$$

$\mathcal{K}$ governs whether a configuration achieves stability. Stable configurations are those for which $\mathcal{K}$ reaches a local extremum — a minimum or maximum in the combinatorial landscape of valid arrangements. These are the substrate analogs of bound states. All observed stable particles are candidates for such extrema.

Level 3 — The Hamiltonian. Under projection into the observer's spacetime coordinate system, $\mathcal{K}$ becomes $\hat{H}$, the Hamiltonian of standard quantum mechanics:

$$\hat{H} = \mathrm{proj}(\mathcal{K}) \tag{6}$$

The Hamiltonian is not an assumption of the theory. It is a derived consequence of the substrate structure, projected through the observational boundary.

The Adjacency Constraint. Two relational nodes are adjacent if their co-manifestation is direct — no third ι event mediates the relation. Valid combinations must satisfy: two ι events of identical distinction character cannot be directly co-manifested. This is not a physical rule but a logical necessity — identical distinction characters in direct co-manifestation produce no new distinction and therefore no new structure. The constraint sharply restricts the combinatorial space and is the source of the minimum complexity threshold below which stable configurations cannot form.

3.7 The Physical Character of ι

The ι event is not point-like. A point has no internal structure. A point-like ι could vary in magnitude or type but not in kind.

ι has multiple relational modes. Not the binary $\{+1, -1\}$ of the field — those are the distinction characters produced at encounter. The relational modes are qualitatively distinct aspects of the ι event itself. Not binary opposition. Not a single axis. Multiple distinct modes — as quark color charge is not magnitude but three qualitatively distinct states that combine under specific rules. The number of modes is not assumed. It is constrained by what the framework must produce. This connects directly to the open question in Unresolved Questions: whether $\{+1, -1\}$ is the full description of ι's internal structure, or whether the relational modes require richer characterization.

ι has intrinsic cohesive attraction. Not because it is complementary to another ι event. Not because it is missing something an opposite supplies. Attraction is what ι is. A single ι event in isolation has this property. No second event is required to activate it. The attraction is mode-dependent — different relational modes produce different kinds of attraction, not merely different magnitudes.

From intrinsic attraction, self-organization follows without instruction. ι events draw toward each other. The arrangement falls into the lowest-energy configuration the relational modes permit. The geometry emerges from the constraint — not assumed, not designed. In three-dimensional projection this corresponds to the class of energy-minimizing space-filling structures: the arrangements nature defaults to wherever energy seeks equilibrium under packing constraints.

No packing process is perfect. At the ι scale, stochastic fluctuations are inevitable. Some regions configure slightly differently than others. A relational mode misaligned. An arrangement slightly off. These imperfections are guaranteed by probability alone.

The imperfections persist and propagate. A local misalignment strains neighboring ι events. They adjust. The adjustment strains their neighbors. The disturbance propagates outward through the medium.

> Mass is a stable stochastic imperfection in the ι configuration.

A persistent local deviation from the ground-state arrangement that the surrounding medium continuously attempts to correct.

> Gravity is the strain field propagating through the medium as it attempts to correct toward ground state.

Not a force transmitted across a void. Not a signal. The medium's continuous attempt at restoration, propagating through the ι network itself. Infinite reach follows without residue. Every ι event participates in the ground state. Every stable imperfection propagates strain outward through all ι events. No carrier particle required. The participation is constitutive.

Status: Candidate direction developed 2026-05-23. Requires formal development.

3.8 The Substrate — Complete

The substrate description is now complete. It contains:

- One substrate: the Medium
- One organizing claim: the independence primitive — processes occur independently; existence may impact existence
- One property: ι — the Medium's act of self-distinction, producing character as the difference the distinction makes
- One field: $\phi \in \{+1, -1\}$ — a record of distinction characters
- One relationship: co-manifestation $x \sim y$
- One algebraic hierarchy: $\iota \to \kappa \to \mathcal{K} \to \hat{H}$
- One arc: maximum undifferentiation $\to$ structure $\to$ maximum undifferentiation

  Nothing else belongs here. Everything else is downstream.

4 PART II: THE VIEW LAYER

Everything in Part II is downstream of the substrate. Dimensions, constants, observers, and all of physics live here — valid, precise, and complete as descriptions of projected structure. Not descriptions of the substrate itself.

4.1 The Three-Level Structure

The framework has three levels, not two.

Level 1 — The Substrate. ι events. The Medium distinguishing within itself. Raw distinction — +1 or 1 — occurring without pattern, without structure, without dimensionality. The ϕ field records the distinction characters produced. The adjacency constraint is the one logical rule that applies here. Nothing else belongs at this level.

Level 2 — The Relational Web. Registered ι distinction characters organized by the adjacency constraint into co-manifestation relations. Stable configurations where $\mathcal{K}$ reaches a local extremum. This level is structured but pre-dimensional.

The relational web is stochastic. Not deterministic. Not programmed. Stochastic all the way down — from the substrate through the relational web. Probability is the actual character of both levels, not a limitation of knowledge.

The relational web is the probability landscape itself. The accumulated weight of stochastic ι interactions creates regions of higher and lower likelihood. No mechanism required. Statistics operating at scale. Like temperature emerging from molecular motion: no single molecule has temperature, but the aggregate of molecular motion is temperature. The relational web does to ι interactions what temperature does to molecular motion — it collapses granular activity into a readable signal.

The relational web has structural consequence without active intervention. Accumulated pattern shapes what comes next — not because anything decides, but because the worn path makes that route more likely. The aggregate feeds back into the substrate purely by existing.

What the view layer reads as a particle, a mass, a field is a region of high probability density in that landscape — stable enough, persistent enough, dense enough to resolve into something nameable. Stochastic imperfections that are stable become mass. Transient fluctuations cancel in the aggregate and wash out.

Level 3 — The View Layer. The relational web as experienced by an embedded observer. The observer is itself a stable Planck bit configuration complex enough to sustain a systematic account of what the relational web presents. That account takes the form of a dimensional coordinate system: length, time, mass, charge. The stable configurations the relational web presents are read as particles. The relational geometry is read as spacetime. The constants of physics — c, ι_{proj}, G — are the seams where the observer's dimensional framework meets the pre-dimensional relational web.

The view layer is where physics operates. It is valid, precise, and complete as a description of what the relational web presents to an observer. It is not a description of the relational web itself, and not a description of the substrate.

This three-level structure resolves several questions the two-level account left open. Why do particles have the properties they do? Because stable Planck bit configurations at $\mathcal{K}$ extrema have those properties at the relational web level — the observer reads them as mass, charge, and spin at the view layer. Why does high-energy collision not reliably reveal substrate structure? Because collisions operate at the view layer — smashing view-layer particles together and reading the debris. The debris consists of view-layer forced states resolving back to what the relational web stably supports. The relational web level is not directly accessible to view-layer observation.

Status: Established in v0.5.

4.2 The GR/QM Dissolution

General relativity and quantum mechanics have resisted unification for a century. Every attempt to quantize gravity fails. Every attempt to embed quantum mechanics in curved spacetime breaks down at the Planck scale. The conflict is treated as a technical problem awaiting the right mathematical framework.

It is not a technical problem. It is a category error.

GR and QM are not competing descriptions of the same thing. They are accurate descriptions of different layers.

The substrate is genuinely granular. Discrete ι events. Discrete interactions. No determinism. Probability is the actual character of the substrate, not a limitation of knowledge about it.

The relational web is the probability landscape of those stochastic interactions. Still granular in origin. Still stochastic. But accumulated over massive numbers of simultaneous ι events, producing regions of density.

The view layer is smooth. The averaging process produces continuity. GR's spacetime manifold is real — at the view layer. It is what the relational web's probability landscape looks like to an embedded observer constructing a dimensional coordinate account.

Quantum mechanics is substrate granularity leaking through to the view layer. The wave function is the view layer's representation of the probability landscape the relational web produces. The Born rule — probability proportional to amplitude squared — is how the view layer reads probability density. Wave function collapse is not a physical event. It is the view layer resolving one reading from a probability landscape that was never definite.

General relativity is the smooth aggregate. Spacetime curvature is the view-layer account of large-scale probability density gradients in the relational web.

> Quantizing gravity fails because gravity is already the averaged-out result. You cannot granularize something that only exists because granularity was averaged away.

The mathematical failure is the signal. Both frameworks are correct. They are correct at different levels. The conflict between them is the conflict of levels mistaken for a conflict of theories.

Status: Candidate dissolution developed 2026-05-23. Requires formal development.

4.3 The Observer's Position

Before describing the projections, it is necessary to name what the observer is and what it is not.

An observer is a self-sustaining structure in the binary field — a stable pattern that persists across transitions, embedded in the differentiated region it constitutes. Every observer who has ever done physics is such a structure. Every human observer is an earth-bound variant: carbon-based, temperature-sensitive, confined to a narrow spatial and temporal window, equipped with sensory apparatus tuned to a small fraction of the available signal, and dependent on cognitive architecture shaped by survival pressures that predate science by orders of magnitude.

These are not complaints. They are structural facts that bear on what physics can and cannot see from its current position.

The observer arrives late. The substrate has been operating — in whatever sense "operating" applies to an eternal, timeless ground — without observers for the overwhelming majority of the arc. The observer arrives embedded in a mature, highly differentiated region of the binary field. It has no access to the substrate prior to structure formation. It has no access to regions beyond the propagation horizon of its own local structure. Its coordinate system — meters, seconds, joules, kelvins — was built to make *its own scale* tractable. That coordinate system is extraordinarily powerful within its domain. It is not a description of the substrate. It is a description of the projection.

The constants of physics — h, c, G, k — are the seams where the observer's dimensional framework meets the substrate's dimensionless operations. They are not brute inputs. They are artifacts of the position from which the measurement is taken.

A different observer, embedded in a different region of the binary field at a different stage of the arc, with a different sensory range and a different coordinate system, would derive different constants and call them fundamental. They would be equally right and equally downstream.

4.4 Physics as Projection

The Medium is pre-dimensional. The binary field ϕ carries no units. The transition index t is a dimensionless label. c does not exist at the substrate level. The Independence Primitive establishes that processes occur independently with no shared temporal framework. A rate requires counting transitions against time. No substrate-level time exists. Therefore no substrate-level rate exists, and c as a transition rate limit is a category error at the substrate level. c is view-layer emergent — the ratio of projected distance to projected time, fixed by the structure of projection itself.

> Dimensions do not exist in the substrate. They are constructed by observers.

To describe the relationships between structures like itself, the observer constructs a coordinate system — a framework of units and dimensions that makes the relationships tractable. That coordinate system is not a discovery about the substrate. It is a construction of the observer.

> Every dimensional quantity in physics — every constant, every observable, every equation — is a projection of a substrate relationship onto the observer's dimensional coordinate system. Physics is the complete and consistent description of those projections. It is a perfect map of the shadow. It is not a description of the substrate that casts it.

4.5 The Projection Boundary

The Planck scale marks the projection boundary: the threshold at which one ι event becomes visible to the observer's dimensional framework.

In the Medium's unit system, one ι event is 1. Not 1 of something. Just 1 — the irreducible unit of occurrence. No further specification is needed or possible.

What physics calls Planck's constant does not exist as a fundamental constant in the Medium. $\hbar = 1.054 \times 10^{-34}$ J·s is the SI measurement of 1 ι event projected — the size of the projection boundary expressed in a unit system built from human-scale standards. That specific number is a statement about the relationship between human-scale SI units and the projection boundary. It is not a fact about the substrate. In Medium units it is simply 1.

The constancy of $\hbar$ in SI follows from this directly. Every ι event is identical — the same occurrence of the same property of the same substrate. The observer always measures the same thing and always gets the same number. The constancy is not a law of physics. It is a consequence of 1 always equaling 1.

An open question is carried explicitly: why does the projection carry what the observer calls units of action — energy multiplied by time — and not some other dimensional combination? The answer requires a derivation of the observer's dimensional categories from the structure of sustained binary field configurations. That derivation is a primary target for future development.

4.6 Natural Units of the Framework

SI units are not available to the Medium as a foundation. They were constructed by observers embedded in human-scale experience — built from the meter, the kilogram, the second, and the ampere, each defined by reference to phenomena accessible at that scale. They carry the fingerprints of the observer's position. To express substrate quantities in SI is to import the dimensional categories of the view layer into the pre-dimensional level. That is the inherited error the framework is attempting to correct.

Planck units are not an alternative. They are SI constants rearranged — $\ell_P = \sqrt{\hbar G/c^3}$, $t_P = \sqrt{\hbar G/c^5}$, $m_P = \sqrt{\hbar c/G}$ — still derived from $\hbar$, c, and G, which are themselves observer-measured quantities. Planck units rescale the observer's framework. They do not replace it.

The Medium requires units built from the substrate upward.

Substrate level. The primitive unit is the ι event — one act of self-distinction. It is the irreducible quantum of occurrence. No smaller unit exists because no smaller event exists. The substrate has no dimensions. ι events are counts. Characters are elements of $\{+1, -1\}$. κ and $\mathcal{K}$ are dimensionless integers. All substrate quantities are pure numbers.

Projection boundary. When a ι event crosses the projection boundary, the observer's dimensional framework assigns it a measurable quantity. In the Medium's unit system, one ι event $= 1$. This is the irreducible unit of the framework. It requires no further definition.

What SI physics calls Planck's constant $\hbar$ does not exist as a fundamental constant. It is the SI numerical value of 1 ι event projected — the size of the projection boundary expressed in human-scale units. The specific value 1.054×10^{-34} J·s is a unit conversion artifact, not a fact about nature. In Medium units it is 1. The Medium's unit of action is:

$$[\mathcal{A}] \equiv \iota_{\text{proj}} = 1 \tag{7}$$

The SI measurement $\hbar$ locates the projection boundary on the SI scale. It does not define the unit. The framework defines the unit. The SI value is verification.

Derived projection units. From $[\mathcal{A}]$ and the structure of projection, minimum units of length, time, mass, and charge are derived at the projection boundary. These coincide empirically with what SI physics calls the Planck scale — but they are defined from the substrate, not from SI constants. Their numerical values in SI are verification, not foundation.

Going forward. Substrate quantities are expressed as pure dimensionless numbers. Projection-layer quantities are expressed in terms of ι_{proj} and its derived units. SI values appear only when comparing derived results to experiment, and are clearly marked as empirical correspondences.

The derivation of the projection units from substrate structure — why one projected occurrence carries the specific action value it does, and why the minimum projected length, time, and mass take the values they do — is an open derivation carried in Unresolved Questions. The unit system is established here as a requirement of the framework. Its full justification awaits that derivation.

Status: Established in v0.5.

4.7 Emergent Properties

The Medium distinguishes three categories of emergent property, organized by the level at which they arise.

Substrate emergent. These properties arise directly from ι-combinations, prior to any observer: mass (corresponding to N, the count of ι events in a configuration), charge (corresponding to the net distinction character of $\mathcal{K}$), spin (arising from relational closure of the configuration's encounter pattern within the web), the quantum vacuum (sub-threshold substrate combinations, appearing at the view layer as vacuum fluctuations — measurable consequences such as the Casimir effect and Lamb shift are open derivations), the Hamiltonian (as $\text{proj}(\mathcal{K})$), spacetime (as the projection framework constructed by embedded observers), and dimensionality (as the coordinate structure observers impose on substrate relationships).

View layer emergent. These properties arise from interactions between stable configurations within the spacetime projection: the fundamental forces, quantum fields, and the full complexity of observable physical reality. This is the domain of existing physics. The Medium makes no claims in this domain beyond identifying it as downstream.

Unassigned. The following have not yet been placed with confidence in either layer: color charge, gravity, neutrino oscillation, and dark matter. Dark energy is not unassigned — it dissolves as a projection artifact of misread redshift. Gravity is the most consequential unassigned phenomenon — if spacetime is itself a projection from ι-relations, gravity may bridge both layers in a way the other forces do not. The current limit of assignment reflects the limit of present understanding, not the limit of The Medium.

4.8 Mass, Charge, and Spin as Projection Signatures

Standard physics treats mass, charge, and spin as independent intrinsic properties of particles. The Medium identifies them as three projection signatures of a single underlying combinatorial structure.

Mass corresponds to N as a first approximation: the total count of ι events in a stable configuration. A heavier particle is a configuration with a larger N. The specific values — why the muon is 207 times the electron's mass, why three generations exist at all — are not yet derived from substrate structure.

Charge corresponds to the net distinction character of $\mathcal{K}$. A configuration in which one distinction character dominates projects to positive charge; the opposing character projects to negative charge; exact balance projects to charge neutrality. Charge is not a property added to a particle. It is the distinction character of its substrate configuration.

Spin describes how a stable configuration's encounter pattern closes within the relational web. It is not rotation. The particle does not spin. The word is retained from the observation — discrete deflection in Stern-Gerlach — but the property is relational, not mechanical.

A configuration whose encounter pattern returns to its original relational state after one traversal of the web projects as integer spin. A configuration requiring two traversals projects as half-integer spin. The distinction is not internal to the configuration alone. It is in how the configuration's encounter pattern sits within the surrounding web.

The specific derivation of which encounter patterns produce which spin values (0, 1/2, 1, 3/2, 2) remains an open problem. The claim that spin is a property of relational closure is current.

4.9 Observer and Observed

Standard quantum mechanics imports a Newtonian posture. A detached external entity measuring a passive system. That posture breaks down at the quantum scale. It was never properly grounded anywhere.

In The Medium there is no outside. Every interaction is between -level processes within the Medium.

The observed is any process whose boundary state is altered by interaction. The observer is any process that retains a structural consequence of that interaction registration. The asymmetry dissolves. Registration is mutual and structural. Not performed from a privileged external position.

A note on vocabulary. The word observation carries two incompatible meanings that the framework must keep distinct. In QM, observation means the interaction that produces a definite outcome no consciousness required, purely structural. In ordinary language, observation means the experience of a perceiving entity. The Medium uses registration for the first and observer for the second. Registration is the structural event at the projection boundary. The observer is the Level 3 embedded configuration constructing a dimensional account of what the relational web presents. These are different things at different levels. Conflating them is the source of a century of confusion about the measurement problem.

Superposition is not a rapid toggling between states interference patterns from the double slit rule that out definitively. It is the formalism's acknowledgment that before interaction, definite states do not exist. The measurement problem has gone unsolved for a century because the observer/observed distinction was never grounded. The Medium dissolves the distinction at the substrate level. What physics calls measurement is registration. What physics calls observer is any process capable of retaining a structural consequence of boundary contact.

4.10 Co-Manifestation and Entanglement

At the substrate level, two ι events producing the same ϕ character share a state with no spatial relationship between them. There is no distance at the substrate level. There is no *between*.

When an observer embedded in differentiated structure encounters two events that are correlated regardless of their spatial separation, it calls this entanglement. Entanglement is the observer's projection of co-manifestation $x \sim y$ into a framework that expects spatial locality to govern correlation. The apparent mystery of entanglement — correlation without causal connection across space — dissolves at the substrate level, where space does not yet exist and co-manifestation requires no connection at all.

4.11 The Stochastic Character — Observer's Description

At the substrate level, ι events are not driven. No prior state determines which relational node the Medium next distinguishes within itself. Each act of self-distinction produces a character — +1 or -1 — with no preference and no guarantee. An observer describing this stochastic character mathematically recognizes it as a Bernoulli process — a binary outcome with associated probability $p(\iota)$ for each ι event. This is the observer's mathematical framework imposed on the substrate's stochastic character. The substrate does not know about Bernoulli. Distinction simply occurs or does not.

The full statistical mechanics of the ground state — entropy, temperature, distributions — emerges on the observer side as the projection of the substrate's undifferentiated stochastic character into the observer's thermodynamic framework.

4.12 The General Principle

The presence of dimensional constants in the equations of physics — h, c, G, k — is the signature of projection. Each constant marks a place where the observer's dimensional framework has been imposed on a substrate relationship that has none. In the substrate's own terms, those constants do not appear.

The program implied by the symbol table: for each row relating a substrate quantity to an observer quantity, derive the dimensional form of the observer quantity from the substrate quantity and a formal account of the projection boundary. If that program succeeds, every fundamental constant of physics will have been given a substrate derivation — and the mystery of their values resolved, not by finding deeper constants, but by understanding that they were never fundamental to begin with.

4.13 On Existing Physics

The existing mathematical frameworks of physics are valid descriptions of differentiated structure behavior within their respective domains. They are not wrong. They are downstream. They are precise and consistent descriptions of the observer's projection of the substrate.

The Medium framework does not replace them. It identifies the substrate from which they project — and defines the program by which their dimensional constants can, in principle, be derived rather than assumed.

Three features of existing physics that appear to be gaps in The Medium are not gaps. They are artifacts of the mathematical starting point those frameworks adopted.

The complex structure of quantum mechanics. Quantum mechanics requires complex-valued wave functions — amplitudes are complex numbers, the Schrödinger equation contains an explicit imaginary unit. This was not derived from first principles. It was adopted: Schrödinger borrowed wave mathematics from classical physics; the complex Hilbert space formulation was chosen because it worked. The Medium does not need to derive complex numbers. It needs to derive the phenomena — interference, superposition, the uncertainty relations — from substrate structure. If those phenomena emerge from co-manifestation and relational closure, the observer will find that complex Hilbert space is a useful mathematical description of what they are seeing. That is the description, not the ground. The complex structure is a view-layer formalism built on a wrong starting point. The phenomena are real. The mathematics is downstream.

Renormalization. Quantum field theory on continuous spacetime produces infinities in its calculations. These are absorbed into redefined parameters through renormalization — a mathematically consistent but physically unsatisfying procedure. The infinities arise because the theory assumes no minimum length scale: arbitrarily short wavelengths contribute to loop integrals. The Medium has a minimum scale: one ι event is the irreducible unit of distinction. Below that there is nothing to integrate over. The UV divergences that require renormalization do not arise. The problem dissolves because the substrate removes the assumption that generated it.

The Schrödinger equation. If the Hamiltonian is derived as $\mathrm{proj}(\mathcal{K})$, and time is view-layer emergent, then the Schrödinger equation $i\hbar\partial\psi/\partial t = \hat{H}\psi$ is a view-layer description — the observer's account of how configurations evolve in projected time under the derived Hamiltonian. It is not a substrate input. It is downstream of the substrate structure from which the Hamiltonian emerges.

The uncertainty principle. Heisenberg's uncertainty principle — $\Delta x \Delta p \geq \hbar/2$ — is often presented as a fundamental postulate. It is not. It is a mathematical consequence of Fourier transforms: a wave with perfectly definite frequency extends infinitely in space; localizing a wave requires superposing many frequencies; precise position means indefinite momentum and vice versa. That is a mathematical relationship, not a physical constraint imposed from outside. In The Medium, position and momentum are both view-layer constructs. Neither exists at the substrate level. When the observer projects substrate structure into a coordinate system and assigns position and momentum, the Fourier relationship between the two conjugate constructs is inherited from the wave structure of the projection itself. The uncertainty principle follows automatically. It is not a substrate feature requiring separate derivation. The stochastic character of ι events — no prior state determines the next — is the substrate ground of the uncertainty. The specific form $\Delta x \Delta p \geq \hbar/2$ is the view-layer expression of that ground, shaped by the Fourier relationship between the position and momentum bases the observer constructs.

The information paradox. Hawking showed that black holes emit thermal radiation carrying no information about infalling matter. If a black hole evaporates completely, information appears to be destroyed. This contradicts quantum mechanical unitarity — unitary time evolution requires information to be conserved. The paradox has been central to theoretical physics for fifty years. The Medium dissolves it. Unitarity is a property of the Schrödinger equation. The Schrödinger equation is view-layer — the observer's account of how configurations evolve in projected time. It is not a substrate input. The substrate is stochastic: no prior state determines the next ι event. There is no substrate-level requirement for deterministic reversibility. Unitarity is a view-layer property that holds for isolated systems within the view layer. When a black hole forms and evaporates, the configurations that produced it dissolve back into unconstrained ι events. From the view layer this looks like information loss. From the substrate it is the stochastic ground returning to its undifferentiated state. The paradox is generated by applying a view-layer constraint — unitarity — to a process that passes through the substrate. That constraint does not apply there. The paradox dissolves.

The strong CP problem. QCD has a parameter θ that can range from 0 to 2π. No theoretical reason constrains it. The natural value is order 1. Experiments show $\theta < 10^{-10}$ — essentially zero. The strong force does not violate CP symmetry even though the theory permits it. The standard proposed solution introduces a new symmetry and a new particle (the axion) to dynamically drive θ to zero. The Medium dissolves the problem. The parity violation account in Unresolved Questions argues that CP violation at the view layer arises when a time direction is established — specifically for configurations with half-integer relational closure, through the topos mechanism. The weak force couples to chirality, which requires a time-directed context. The strong force couples to color charge — a relational labeling of three-quark closure with no directional preference. SU(3) color is the freedom to relabel color assignments without changing physics. That relabeling freedom has no time-direction component. Projecting through a time-direction-neutral channel produces no CP violation. $\theta = 0$ is the natural result of a parity-symmetric substrate projecting through a coupling channel that does not engage the time-direction mechanism that breaks symmetry in the weak sector. No axion required. The strong force is CP-symmetric because its projection channel is direction-neutral.

Bell's theorem. Bell's theorem proves that no local hidden variable theory can reproduce quantum mechanical predictions. Experiments confirm quantum mechanics. Local hidden variable theories are ruled out. The Medium's substrate may appear to be a hidden variable theory — ι events are below the observational floor, not directly measured, determining the characters that project to observable physics. That pattern is what hidden variable means. The Medium is not a local hidden variable theory. Bell's theorem requires locality: influences cannot propagate faster than c between spacelike-separated events. The Medium's substrate has no c — c is view-layer emergent. The substrate has no spatial separation — space is view-layer. Co-manifestation is explicitly non-local: two ι events sharing a state with no spatial relationship between them. There is no distance at the substrate level. Bell's locality assumption does not hold at the substrate level. The theorem's premises do not apply. The Medium is a non-local substrate theory. The locality that Bell requires is a view-layer artifact that does not exist at the ground floor. The apparent mystery of entanglement — correlation without causal connection across space — is not mysterious at the substrate level. Space does not yet exist there. Co-manifestation requires no connection across it.

5 PART III: COSMOLOGICAL IMPLICATIONS

The following develops four interconnected consequences that follow directly from the foundational structure of The Medium. Each addresses a major puzzle in standard cosmology. In each case the puzzle dissolves rather than being solved — it was generated by a category error that The Medium does not commit: treating view-layer artifacts as substrate-level facts.

5.1 Entropy as Emergent Accounting

Standard cosmology inherits a puzzle from Boltzmann: if the second law of thermodynamics is statistical rather than fundamental, why was entropy low at the beginning? The standard response bundles this into initial conditions and calls it the past hypothesis. The Big Bang is asserted to have been an extraordinarily low-entropy state. This assertion is not derived. It is required by the model and inserted by hand.

The Medium does not accept this framing, and not merely because it rejects the Big Bang as an explanation. The more fundamental problem is that the framing assigns entropy as a global intrinsic property of the universe — a quantity that belongs to the cosmos as a whole and increases over cosmic time. This is a category error.

Entropy, as Boltzmann formalized it and Shannon made fully explicit, is a measure of an observer's ignorance about the microstate of a system given knowledge of its macrostate. It is not a property of the system alone. It is a property of the relationship between an observer with limited resolution and a system with more degrees of freedom than the observer can track. Entropy is always relative to a coarse-graining — a choice, made by an observer, of which distinctions count and which do not.

The universe as a whole has no external observer. Assigning it a global entropy value requires an observer standing outside it with a complete accounting of all microstates, which is incoherent. What cosmologists call the entropy of the universe is always the entropy of the observable universe as seen from a particular embedded vantage point with particular resolution limits. It is local. It is relational. It is view-layer all the way down.

In The Medium, the substrate — the ι-combinatorial structure — has no thermodynamic direction. There is no privileged low-entropy initial condition because there is no initial condition at the substrate level in any cosmologically meaningful sense. The apparent entropy gradient that standard cosmology treats as a cosmic fact is a view-layer phenomenon: an artifact of embedded observers constructing temporal sequences and tracking accessible macrostates from within a particular slice of configuration space.

Entropy is therefore classified in The Medium as view-layer emergent. It has no substrate analog.

5.2 Time as Projection and Its Consequences for Frequency

The Medium establishes that time is not a substrate-level property. Temporal directionality — the sense that events occur in sequence, that causes precede effects, that entropy increases in one direction — emerges at the view layer as observers at a particular scale construct ordered accounts of substrate transitions.

Time is real for the embedded observer. Thermodynamically, experientially, irreversibly real. Dismissing it as illusion is philosophical evasion. But real-for-the-observer and necessary-for-the-universe are different claims entirely. The universe's processes do not require time to coordinate. Gravity does not consult a clock. The gravitational relationship between earth and moon is not a sequence of instructions passed back and forth — it is a continuous configuration of the substrate, of which the 1.3-second propagation delay at the view layer is the observer's local account. Physics wrote the observer's experience of sequence into the equations as a global feature of reality. The equations keep resisting it. General relativity is time-symmetric. The Wheeler-DeWitt equation has no time variable. Time is emergent downstream of process. Not upstream of it.

This has a direct consequence for the concept of frequency. Frequency is defined as cycles per unit time. If time is view-layer emergent, then frequency is also view-layer emergent. It is not a substrate-level property of a propagating pattern. It is a measurement that an observer constructs by counting transition events against a locally projected time reference.

Two observers projecting different temporal frames — because they sit at different configuration depths within the substrate, or because their local ι-configuration densities differ — will assign different frequency values to the same arriving pattern. This is not a Doppler effect. It is not recession. It is a structural consequence of time being a local projection rather than a universal background.

5.3 Redshift as Multi-Layer Projection Artifact

Standard cosmology interprets cosmological redshift as evidence of recession velocity: distant galaxies are moving away, space between them and us is expanding, and their light is stretched to longer wavelengths in proportion to the expansion. This interpretation underlies the Big Bang model, Hubble's law, and the entire framework of an expanding universe with a singular low-entropy origin.

The Medium offers a more parsimonious account that requires no expansion, no recession, and no singular origin — while accepting the observational data entirely.

Within The Medium, what we call a photon is a stable propagating ι-configuration pattern — a structure that persists and moves through successive co-manifestation relationships across configuration space. What we call speed is a view-layer quantity: a ratio of projected distance to projected time constructed by the observer. It is not a geometric velocity through space. It is not a substrate-level rate. The substrate has no rates — no shared temporal framework exists against which transitions could be counted.

Redshift arises from three stacked projection effects.

First: the emitting configuration. A source at some region of the substrate produces a transition pattern at a rate determined by local ι-configuration density. That rate is what the source's local view layer calls frequency.

Second: traversal through configuration depth. The pattern propagates across substrate regions that are not uniform. Configuration density varies across the substrate — not directionally, not as expansion, but structurally, as a feature of the combinatorial landscape. As the pattern traverses this varying density, the relationship between transition events and projected distance is not constant.

Third: the receiving configuration. The observer receives the arriving pattern and measures its frequency against a locally projected time reference. That time reference is constructed from the observer's local ι-configuration density, which differs from the emitter's. The frequency the observer assigns is a ratio constructed entirely within the view layer, against a local clock that has no substrate-level universality.

The redshift that results from this three-layer process is not evidence that the source is receding. It is evidence that the emitting and receiving configurations sit at different depths within a non-uniform substrate, and that the projection relationship between them is asymmetric. What standard cosmology interprets as Hubble's law — a proportional relationship between distance and recession velocity — is, in The Medium, a projection artifact of sampling along a particular observational cone from a single embedded vantage point through a substrate with a particular large-scale configuration structure.

The mature and fully structured galaxies observed at high redshift — including findings from JWST that are unexpected under standard galaxy formation timelines — are consistent with this account. They do not violate a law of nature. They strain the galaxy formation models layered on top of the Big Bang framework, not the framework's core timeline. In The Medium, structural complexity at any configuration depth is not constrained by elapsed cosmic time, because cosmic time is itself a projection.

Dark energy dissolves in this account. It was introduced in 1998 to explain why Type Ia supernova redshifts were slightly larger than expected for a decelerating universe — the data indicated accelerating expansion and dark energy was inserted to drive it. It was not derived. It was a fix. If redshift is configuration-depth asymmetry rather than recession, there is no expansion to accelerate. If there is no accelerating expansion, dark energy as a concept has no referent. The cosmological constant problem — why Λ is 120 orders of magnitude smaller than QFT vacuum energy predictions — also dissolves in this form, because it presupposes Λ must be nonzero. What remains is a narrower question: why do sub-threshold substrate fluctuations not gravitationally dominate the view layer? That question is carried in Unresolved Questions. Dark energy as a phenomenon is not.

5.4 The Big Bang as Projection Error

The Big Bang model rests on three foundational assumptions, each of which The Medium addresses directly.

First: that cosmological redshift indicates recession velocity, implying an expanding universe with a singular dense origin. Second: that entropy has a global cosmic direction, requiring a special low-entropy initial state. Third: that time provides a universal background against which an origin event — a moment of creation — is a coherent concept.

The Medium has shown that all three assumptions are view-layer artifacts. Redshift is configuration-depth asymmetry misread as velocity. Entropy is a local observer relationship with no substrate reality. Time is a local projection with no substrate universality.

Remove those three assumptions and the interpretation built on them loses its foundation. The observational data remain — including the cosmic microwave background and the abundance ratios of hydrogen, helium, deuterium, and lithium predicted by Big Bang nucleosynthesis. The Medium does not yet account for those. They are carried as open problems in the Unresolved Questions. What does not survive is the interpretive framework erected on top of the data — the claim that redshift requires recession, that entropy requires a special initial state, that time requires a singular origin.

This is a stronger position than disproof. Disproof engages a model on its own terms, accepting its conceptual foundations and showing internal inconsistency. What The Medium establishes is more fundamental: the conceptual foundations were never valid. The Big Bang is not a theory that turned out to be wrong. It is a theory that was never well-formed — a projection error that becomes visible the moment one insists on asking what is actually happening at the substrate level.

The universe, in The Medium, has no beginning in any substrate-meaningful sense. It has no privileged center, no singular origin, no global clock, and no cosmic entropy gradient. What it has is a substrate of extraordinary combinatorial richness from which structure, observers, and the local projections we call time, space, and thermodynamics all emerge — consistently, without residue, and without requiring any special initial conditions.

The Borde-Guth-Vilenkin theorem holds that any spacetime with an average positive expansion rate must have had a past boundary — a beginning. The theorem is general enough to apply even to inflationary cosmologies. The Medium's response: expansion is a view-layer artifact. The substrate has no expansion rate. The premises of BGV do not hold at the substrate level, and the theorem therefore does not apply there. Whether projection recovers a structure to which BGV would apply is an open question carried in Unresolved Questions.

5.5 The Flatness Problem as Projection Artifact

Standard cosmology requires the universe to be extraordinarily flat — the energy density lies within one part in 10^{60} of the critical density in the early universe. Flatness is unstable under standard expansion: small deviations grow over time. For the universe to be this flat today, initial conditions had to be extraordinarily fine-tuned. Inflation was introduced partly to solve this — exponential expansion drives any initial curvature to negligible levels.

The Medium dissolves the problem without inflation.

Space is view-layer emergent. The geometry of space is constructed by embedded observers from the relational web of co-manifestation. There is no pre-existing spatial geometry to be curved or flat. The relational web has a large-scale structure determined by the statistical properties of co-manifestation across the substrate. The substrate has no preferred direction — direction is view-layer. The projected geometry therefore inherits no preferred curvature. Flatness is not a fine-tuned initial condition. It is the generic result of projecting a substrate with no preferred spatial orientation. Any large-scale curvature would require a preferred direction in the substrate. No such direction exists.

The flatness problem is generated by assuming space is a pre-existing container that could have started curved. Remove that assumption and the fine-tuning dissolves. The universe is flat because the substrate from which space is projected has no mechanism for producing preferred curvature.

5.6 The Horizon Problem as Projection Artifact

Standard cosmology's horizon problem: regions of the CMB separated by more than about two degrees were outside each other's particle horizon at the time of last scattering. No light signal could have connected them. Yet they have identical temperatures to one part in 100,000. Inflation solves this by expanding a causally connected region to cosmic scales.

The Medium dissolves the uniformity without inflation.

Causal horizons require two things: a speed limit (c) and a time since origin. Both are view-layer in The Medium. c is a ratio of projected distance to projected time. The Big Bang timeline is a projection artifact. No substrate-level causal horizon exists because there is no substrate-level c or elapsed time.

The uniformity of the CMB is the projection of the substrate's statistical homogeneity. The substrate has no preferred location. ι events are statistically uniform across the relational web. That statistical uniformity projects as uniform temperature. No causal contact was required because causality is view-layer. The two apparently disconnected regions were never disconnected at the substrate level — spatial separation does not exist there.

The temperature uniformity dissolves. The anisotropy spectrum — the precise one-in-100,000 fluctuation pattern that seeded large-scale structure — does not dissolve. It requires a substrate account of which configurations produce which fluctuation amplitudes at which angular scales. That account does not exist and is carried in the CMB entry of Unresolved Questions.

6 PART IV: WHAT THE MEDIUM NEEDS TO PROVE

Not a physical theory with testable predictions. Not a mathematical framework with derivational power.

Three things.

First: that the primitives are genuinely primitive. That occurrence, registration, boundary do not smuggle in hidden assumptions. That they do not secretly depend on Newtonian concepts.

Second: that existing frameworks require something like these primitives to be coherent. That the measurement problem, the time problem, the observer confusion all point to a gap sitting exactly where The Medium works. Three frameworks in particular are candidates for derivation from the substrate.

Planck bits are what encounter characters look like to an observer who cannot see below the registration level. Digital physics assumes the bit as primitive. The Medium shows where it comes from. The path from encounter to registered character to bit is the derivation. It is not complete. The direction is clear.

Loop quantum gravity makes spacetime emergent from spin networks — nodes and edges that are atoms of geometry. The Medium's encounter web is already that relational structure. Stable encounter configurations at $\mathcal{K}$ extrema are the nodes. The encounter relationships between them are the edges. LQG's spin foam — the four-dimensional history of spin networks — is the observer's projection of encounter patterns across what it calls time. The derivation would show that spin networks are what the encounter web looks like when an observer imposes a spatial coordinate system on it.

String theory places vibrating strings at the foundation. Different vibration modes correspond to different particles. In The Medium, different encounter configurations at $\mathcal{K}$ extrema correspond to different particles. The correspondence is directionally plausible. String theory presupposes spacetime and requires ten or eleven dimensions. The derivation would need to show those dimensions emerging from encounter structure. That is the longest path of the three.

None of these derivations is complete. All three are named targets. Showing that existing frameworks are derivable from the substrate is the substance of this second criterion.

Third: that the framework is internally consistent and generative. That thinking from these primitives produces coherent consequences. That it opens questions existing frameworks could not formulate.

The model is the 1905 paper on special relativity. Not complex. Precise. Spare. The argument careful. The boundaries clearly stated.

That is achievable. And it is enough.

7 PART V: DERIVATIONS

7.1 Derivation I: Planck Bits from the Substrate

Digital physics — Wheeler's *It from Bit*, Fredkin, Wolfram — claims the bit as primitive. A binary value, 0 or 1, is the most fundamental thing. Everything physical arises from it. No explanation is offered for where the bit comes from. It simply is.

The Medium derives it.

Step 1. ι occurs. The Medium distinguishes within itself. That is the event. It is not driven. No prior state determines it. It simply occurs.

Step 2. The distinction has two aspects: +1 or -1. Not chosen from outside. Not determined in advance. The character of the distinction itself. Without difference there is no distinction. The character is what distinguishing is.

Step 3. The character is registered in ϕ for that ι event. The field records it. One distinction. One binary value.

Step 4. At the projection boundary, one ι event corresponds to one Planck-scale event. This is the threshold at which the substrate's dimensionless distinction acquires the dimensional form the observer calls action. One ι event. One Planck unit. One registration.

Step 5. What the observer records at the Planck scale is one binary value: +1 or -1.

That binary value is the bit.

A Planck bit is not a primitive. It is the registered distinction character of one ι event, as seen from the observer's position. The bit presupposes the act of distinction. Digital physics correctly identifies the information structure of reality. It misidentifies that structure as the ground floor.

Consequences that follow without additional derivation:

The stochastic character of bits — no prior state determines the next — follows directly from ι events not being driven. No prior state determines where or when the Medium next distinguishes within itself.

The minimum size of a Planck bit — information cannot be compressed below one Planck unit — follows from ι being the unit of distinction. Below one distinction event, there is nothing to register. The Planck scale is not arbitrary. It is the projection boundary of the substrate's irreducible unit.

The adjacency constraint explains why directly co-manifested identical bits produce no new information. Two ι events of identical distinction character in direct co-manifestation produce no new distinction. This is not a rule imposed on the bits. It is a logical consequence of what distinction is.

What is not yet derived:

Why one ι event projects to exactly $\hbar$ of action — energy multiplied by time — and not some other dimensional combination. This remains in unresolved questions. The derivation above stands without it.

7.2 Derivation II: Standard Particles as Planck Bit Configurations

The Planck bit derivation has an immediate consequence. If a Planck bit is a registered ι distinction character, then a standard particle is a stable configuration of Planck bits — a pattern of distinction characters that has achieved stability at a $\mathcal{K}$ extremum.

The properties of that particle are what the pattern projects to the observer.

Mass corresponds to N as a first approximation: the count of ι events in the configuration. A heavier particle is a larger pattern. The lightest particles are the threshold cases — the minimum number of Planck bits that can achieve stability under the adjacency constraint. The specific mass values and the three-generation hierarchy are not yet derived.

Charge corresponds to net distinction character. A pattern in which one character dominates projects as charged. Exact balance projects as neutral.

Spin corresponds to relational closure. The pattern either closes in one traversal of the web or two. One traversal: boson. Two traversals: fermion.

The adjacency constraint shapes the particle table.

Identical adjacent distinction characters produce nothing. Only patterns that vary can build. This sharply limits which configurations are possible. The number of stable particles is not arbitrary. It is constrained by which patterns satisfy the adjacency rule. The particle table may be exactly the set of patterns the constraint permits.

The threshold cases:

The electron is the minimum stable configuration with net character imbalance and half-integer closure. Possibly the simplest stable fermion the adjacency constraint allows.

The photon does not sit at a $\mathcal{K}$ extremum. It is a propagating alternation of distinction characters — the adjacency constraint satisfied but no bound state achieved. The observer assigns it speed c — a view-layer ratio, not a substrate property. Why projection fixes that ratio at c and not some other value is an open derivation.

The neutrino approaches the stability threshold from below. Near-zero mass. Balanced character. The minimum fermion-closing configuration.

Quarks cannot satisfy the adjacency constraint alone. Stability requires combination. The structural impossibility of isolated quark stability is a substrate consequence of the adjacency constraint. The energy-dependent behavior of the strong force — asymptotic freedom, running coupling — is a view-layer QCD result. These are distinct claims at different levels.

The open derivation:

Which specific Planck bit patterns — satisfying the adjacency constraint, achieving a $\mathcal{K}$ extremum — correspond to each observed particle? That is the derivation. It begins with the lightest threshold cases and works upward through the particle table. The framework predicts that the table is finite and complete. The adjacency constraint permits exactly the particles that exist. No derivation yet confirms this. It is the primary structural task of the next stage.

8 Unresolved Questions

These are the frontier of the framework, stated precisely and carried explicitly.

Why Binary. The framework takes $\{+1, -1\}$ as the only two distinction characters. The argument: one act of distinction creates one boundary; one boundary has two sides. That is a logical claim, not a derived result. It cannot be supported by appealing to quantum discreteness or Planck-scale physics without circularity — those observations are downstream of the substrate, not prior to it. The open question is whether the logical argument is sufficient: does one act of distinction necessarily produce exactly two values, or is three-valued or continuous distinction coherent? A three-valued substrate would alter the algebraic hierarchy throughout and may produce the SU(3) color structure more directly. A continuous substrate would recover something closer to quantum field theory from the start. Both alternatives are unexplored. The binary axiom is currently the most defensible choice on grounds of minimalism. It is not the only possible choice.

A candidate direction: the ι event is multifaceted, with multiple qualitatively distinct relational modes — not binary opposition but multiple distinct kinds of interaction, as quark color charge is not magnitude but three qualitatively distinct states. The $\{+1, -1\}$ field may record one aspect of ι's distinction character while the relational modes carry richer structure not yet fully characterized. If the mode count is constrained by what the framework must produce — gravity, electromagnetism, the particle table — then the binary axiom is an underdetermination rather than a falsity. The field records distinction. The modes govern interaction. These may be separable claims. This direction is developed in the Physical Character of ι section above and carried forward as an open structural question.

The Vocabulary Boundary. The terms occurrence, registration, and boundary are proposed as genuinely primitive replacements for the inherited vocabulary of observer, measurement, and collapse. Whether they succeed — whether they are themselves free of hidden Newtonian assumptions — is an open question requiring formal examination.

The Topos Theory Connection. operates below Boolean logic. Occurrence as boundary negotiation is not a yes/no event — it is a process whose definiteness is contextual and relational. Topos theory, developed by Grothendieck, allows contextual relational logic where truth values are not simply binary. The Medium arrives at this logical territory from the conceptual direction; topos theory approaches from the mathematical direction. Whether they converge formally is an open derivation.

The Nature of ι. What is intrinsic distinction? The provisional characterization — the property permitting local, spontaneous reduction from maximum undifferentiation in a fully random substrate — is the central open question of the theory.

The Formal Account of the Projection Boundary. How does an observer embedded in the binary field construct a dimensional coordinate system? What determines which dimensional categories emerge? This is the primary task for the next stage of formal development.

Why Does One Manifestation Project to Action? The observer's measurement of one irreducible substrate occurrence produces a quantity with units of action — energy multiplied by time. Why this dimensional combination? Until that derivation exists, the correspondence between one unit of substrate process and h is a named target, not a result.

The Derivation of Projection Units from Substrate Structure. The framework defines its own unit system built from the substrate upward. The primitive unit is the ι event. At the projection boundary, one projected ι event defines one unit of action $[\mathcal{A}] \equiv \iota_{\mathrm{proj}}$. From this, minimum units of length, time, mass, and charge are derived. These coincide empirically with the Planck scale in SI — but the Planck scale is not the definition. It is the verification.

The open derivation has three parts. First: why does one projected ι event carry units of action specifically — energy multiplied by time — and not some other dimensional combination? The observer's dimensional categories must be derived from the structure of sustained binary field configurations. That derivation does not exist. Second: why does projection fix the ratio of projected distance to projected time at the specific value observers measure as c? c is not a substrate property — the Independence Primitive establishes no substrate-level time and therefore no substrate-level rate. c is a view-layer ratio. Why projection fixes it at the value it does is open. Third: the minimum projected mass unit involves the gravitational structure of projection. That derivation cannot be completed until gravity has been placed confidently in the framework.

These three are the load-bearing open derivations of the unit system. Until they exist, the correspondence between ι_{proj} and the SI value of $\hbar$ is an empirical match, not a derived result.

The substrate has no preferred frame. Frames arise only at projection. Lorentz invariance is a property of how projection works. The derivation confirming that projection recovers exact Lorentz invariance does not yet exist.

The Observation Method Problem. The Standard Model's particle table was built primarily from high-energy collision experiments. These experiments force the view layer into configurations above the stability threshold — energy conditions that do not occur in stable physical reality. The particles produced are almost entirely transient, decaying within fractions of a second into stable configurations. The Standard Model catalogs the full debris of those collisions as fundamental particles. The Medium does not accept that catalog without examination.

The framework distinguishes two classes of measurement. The first: naturally stable configurations — the electron, proton, photon, neutrino, and the lightest stable hadrons. These are candidates for genuine $\mathcal{K}$ extrema in the substrate. They persist. They are not products of forced above-threshold conditions. The second: transient collision products — particles that exist only under extreme forced conditions and decay immediately. These are candidates for above-threshold view-layer forced states, not stable substrate configurations. They reveal the path to stability; they are not the stable configurations themselves.

This distinction matters for the derivation program. The Medium is not required to derive every particle the Standard Model catalogues from collider debris. It is required to derive the naturally stable configurations — and then to show why forced above-threshold conditions produce the transient states observed. The Standard Model treats both classes equally as fundamental. The Medium should not.

Neutrino oscillation sits partially in the first class. Solar, atmospheric, and reactor neutrinos are natural. Their oscillation is observed over long baselines, not in collision debris. That makes neutrino oscillation a more reliable guide to substrate structure than most collider-produced phenomena. The PMNS mixing parameters, however, are partly measured in accelerator experiments, and whether those values are substrate properties or measurement-method artifacts remains open.

Particle Configurations. Stable substrate configurations are those where $\mathcal{K}$ reaches a local extremum under the adjacency constraint. The open derivation: identify the minimum stable Planck bit patterns satisfying the adjacency constraint, match them to the naturally stable observed particles, and show why forced above-threshold combinations produce the transient states the Standard Model catalogs as additional particles. The particle table derivation begins with the lightest stable threshold cases. This is the primary structural task of the next stage.

Gauge Symmetry. The Standard Model is built on SU(3)ŒSU(2)ŒU(1) — the symmetry groups that determine the form of the strong, weak, and electromagnetic forces. Gauge symmetry is why the forces have the structure they do, why particles carry specific charges, and why force-carrying particles exist. The Medium places the fundamental forces at the view layer but does not explain where SU(3)ŒSU(2)ŒU(1) comes from. Why these groups? Why not others? A possible substrate path: gauge symmetry is the freedom to redefine fields locally without changing physics. The complex phase freedom of quantum wave functions — and $e^{i\theta}\psi$ represent the same physical state — is U(1) gauge symmetry. SU(2) and SU(3) are larger instances of the same idea. The substrate relational web has its own symmetry structure: the web of co-manifestation relations has no preferred global labeling. That is a form of gauge freedom at the substrate level. Whether projection of that substrate gauge freedom recovers SU(3)ŒSU(2)ŒU(1) specifically is the open question. If it does, The Medium would derive not just the forces but the exact symmetry group structure that governs them — the deepest result the framework could produce. This derivation does not exist. It is named here as the primary structural target beyond the particle table derivation.

Asymptotic Freedom and the Strong Force. The document claims confinement follows from the adjacency constraint — quarks cannot achieve independent stability because no valid Planck bit pattern satisfying the constraint exists for isolated quarks. That is a substrate-level structural claim. Confinement has a complement: asymptotic freedom. At high energies quarks behave as nearly free particles. The strong coupling weakens as energy scale increases and strengthens as it decreases. This energy-dependent behavior is the defining feature of QCD and was experimentally established in deep inelastic scattering. The adjacency constraint is scale-free — a binary logical rule with no energy dependence. It cannot produce a running coupling. The boundary needs to be drawn clearly. The adjacency constraint explains the structural impossibility of isolated quark stability — a substrate result. The energy-dependent behavior of the strong force is a view-layer QCD result — downstream of the substrate. These are different claims at different levels. The document blurs the boundary by implying the adjacency constraint accounts for confinement generally. It accounts for the structural part. The running coupling and asymptotic freedom are view-layer phenomena not yet connected to substrate structure.

The Higgs Mechanism. The Standard Model gives particles mass through coupling to the Higgs field. Coupling strength determines mass. The Higgs boson was discovered in 2012 at 125 GeV — confirmed physics with a measured particle. The Medium says mass corresponds to N as a first approximation and says nothing about the Higgs. That silence is a gap. Three possible placements. First: the Higgs is view-layer — the observer's account of why different substrate configurations have different N counts, with the Higgs field being the view-layer description of that substrate structure. This would make the Higgs a projection artifact of the N structure, but it would need to explain why the Higgs has its specific mass and why coupling strengths match the observed fermion mass spectrum. Second: the Higgs is substrate-emergent — a specific Planck bit configuration that permeates the Medium and whose co-manifestation with other configurations determines their effective N. This preserves the Higgs as a real phenomenon at the substrate level but introduces a new substrate entity not yet in the framework. Third: the placement is genuinely open. The Higgs mechanism is confirmed physics. The Medium does not yet address it. This is the current position.

Neutrino Oscillation. Neutrinos are produced and detected in flavor eigenstates — electron, muon, tau. They propagate in mass eigenstates — three distinct mass values. The two bases are not the same. A neutrino produced as an electron neutrino is a superposition of all three mass eigenstates. As it propagates, the three components evolve at different rates. The superposition rotates. At some distance the electron neutrino has become predominantly a muon neutrino. This is oscillation — measured precisely, with three mixing angles and a CP-violating phase described by the PMNS matrix. The Medium's particle account treats each stable configuration as a specific Planck bit pattern at a $\mathcal{K}$ extremum. Neutrino oscillation requires a configuration to be genuinely indeterminate between patterns during propagation — not because unmeasured, but because mass and flavor bases are structurally distinct and neither is more fundamental than the other. No mechanism for this exists in the current framework. A possible direction: the question "which flavor is this neutrino?" may have no substrate-level answer. Mass eigenstates are the substrate configurations. Flavor eigenstates are local sections over the detection context — what the weak interaction selects when measurement establishes a context. This is the same topos move applied to fractional charge and parity violation: a property that appears global is revealed as a local section. The PMNS mixing angles and CP-violating phase would still require derivation from substrate structure. That derivation does not exist.

The Quantum Vacuum. Sub-threshold substrate combinations appear at the view layer as quantum vacuum fluctuations. The quantum vacuum is not empty — it exerts measurable pressure and perturbs energy levels. The Casimir effect: two conducting plates in vacuum experience an attractive force from vacuum fluctuations, quantitatively confirmed. The Lamb shift: vacuum fluctuations shift hydrogen energy levels by a precisely measured amount, a result that drove the development of QED. Both are view-layer QED results. Whether the substrate fluctuation account produces the correct vacuum pressure and energy perturbation for these measurements is not shown. The account is directionally consistent — sub-threshold combinations are real substrate structure — but the quantitative bridge from substrate to measured Casimir force and Lamb shift does not yet exist.

Adjacency Constraint and Pauli Exclusion. The adjacency constraint operates at the substrate level: two ι events of identical distinction character cannot be directly co-manifested. Pauli exclusion operates at the view layer: no two fermions can occupy the same quantum state. They are at different levels and are not the same rule. The relationship between them is unexamined. The possibility worth pursuing: the adjacency constraint may generate Pauli exclusion under projection. If identical distinction characters cannot be directly co-manifested at the substrate, configurations that project as fermions — half-integer relational closure — may inherit that exclusion at the view layer. That would be a derivation of Pauli exclusion from substrate structure. The gap: the adjacency constraint applies to all ι events regardless of what they project to. Bosons do not have Pauli exclusion. Why the constraint produces exclusion for fermions but not bosons is not shown. Until that is shown, the adjacency constraint and Pauli exclusion are parallel structures at different levels, not the same result at different scales. This is a named candidate derivation, carried as open.

Spin as Relational Closure. Which encounter patterns produce which spin values (0, 1/2, 1, 3/2, 2)? The claim that spin is a property of relational closure — integer spin closing in one traversal of the web, half-integer in two — is current. The derivation of which specific encounter patterns correspond to which values is open. More consequentially: the framework names the topology of spin but does not yet connect it to exchange statistics. Fermions obey Pauli exclusion — no two in the same quantum state. Bosons do not. That difference determines atomic shell structure, the stability of matter, and the behavior of light. The spin-statistics theorem in quantum field theory derives this connection from Lorentz invariance, causality, and positivity of energy. The Medium's relational closure description is consistent with the topology the theorem requires but does not derive the statistics. Doing so requires showing that projection from the substrate recovers Lorentz invariance, from which the spin-statistics theorem follows as a downstream result. That derivation does not yet exist.

Derivation of Prior Frameworks. Three existing frameworks are candidates for derivation from the substrate. Planck bits: the registered encounter character is a bit — the derivation shows where the bit comes from. Loop quantum gravity: the encounter web maps to spin networks; stable configurations map to nodes; encounter relationships map to edges; the derivation shows LQG as the observer's projection of the substrate's relational structure. String theory: different encounter configurations at $\mathcal{K}$ extrema correspond to different particles as string vibration modes correspond to different particles — the derivation requires showing how the encounter structure generates the dimensional space string theory presupposes. Each is an open derivation. Each is a named target.

Fractional Charge. The framework produces integer net distinction characters from ś1 combinations. Quarks carry charge ś1/3 and ś2/3 — never observed in isolation, always confined within hadrons. The hadronic totals are integers; the quark-level values are not.

The appearance of fractional charge in a binary substrate is not a puzzle to be solved by making the fractions work. It is a signal — the same kind of signal dark energy provided. Dark energy appeared to require explanation until the framework asked why it appears at all, found it was a projection artifact of misread redshift, and dissolved it. The same instinct applies here.

A binary substrate produces integers. The appearance of 1/3 means the quark model is describing substrate structure at the wrong level — carving the substrate at the wrong joints, the same way recession velocity was the wrong description of redshift. Quarks as described in the standard model may not be the right substrate entities. The framework should ask what configuration of ι events produces the behavior attributed to quarks. That configuration will have integer distinction characters. The fractional values appear because the standard model imposes its own framework on substrate structure that has no fractional properties.

The open derivation: identify the ι configurations whose behavior the standard model describes as quarks. Show that those configurations carry integer characters. Show why projecting through the standard model's quark framework produces the appearance of 1/3 and 2/3. Confinement — the impossibility of isolating a quark — is then not a separate physical fact but a structural consequence of the same misidentification: outside the hadronic configuration, the entity the standard model calls a quark does not exist as a substrate unit.

The topos move — treating fractional charge as a local section over the hadronic context — is a candidate direction for understanding why the fractions appear and sum correctly. It is not the primary resolution. The primary resolution is the correct substrate-level identification of the configurations the standard model is describing.

The Mass Hierarchy. Mass corresponds to N as a first approximation. That correspondence does not explain why the electron is 0.511 MeV, the muon 105 MeV, and the tau 1,777 MeV — identical structure, identical charge, identical spin, three wildly different masses. It does not explain why there are exactly three generations. The Standard Model inserts these values by hand. The Medium claims mass is derivable from substrate structure — that claim requires showing which combinatorial configurations produce which specific N counts, and why those counts correspond to the observed values. This derivation does not yet exist.

Matter-Antimatter Asymmetry. The observable universe is made almost entirely of matter. Equal amounts of matter and antimatter should have been produced. The Standard Model satisfies the three Sakharov conditions for baryogenesis — CP violation, baryon number violation, departure from thermal equilibrium — but too weakly by many orders of magnitude to account for the observed asymmetry. The Medium's substrate is symmetric in ś1. No intrinsic matter/antimatter preference exists there. A possible path already in the framework: the parity violation account applies here as well. Parity violation arises at projection when a time direction is established — the topos move in the parity entry below. The CPT theorem requires that matter and antimatter are related by simultaneous reversal of charge, parity, and time. If time reversal is a view-layer operation with no substrate analog — because the substrate has no time — then CPT symmetry is a view-layer statement. The time arrow established at projection breaks the ś1 symmetry in the same operation that breaks parity in the weak force. The substrate is symmetric. The asymmetry is a projection artifact of the time direction the observer constructs. Whether this produces the correct magnitude of asymmetry — the observed ratio of roughly one billion photons per baryon — is entirely unshown. The direction is consistent with the existing framework. The derivation does not exist.

Parity Violation. The weak force distinguishes left from right — it couples only to left-handed fermions and right-handed antifermions. The substrate is parity-symmetric: ś1 distinction characters have no intrinsic handedness, and the adjacency constraint treats +1 and 1 identically. That symmetry appears to conflict with the measured asymmetry of the weak force. The resolution may lie in the same topos move already applied to fractional charge. Chirality — left versus right handedness — requires a preferred direction: momentum relative to spin, in a spacetime with a local time arrow. Both spacetime and the time arrow are view-layer in The Medium. No projection context, no chirality. Chirality is a local section over the context "spacetime with a local time direction established" — it does not extend to a global element of the substrate's symmetric structure. Parity violation of the weak force is then the statement that weak coupling is sensitive to that local section specifically. The substrate is parity-symmetric. The asymmetry arises at projection when fermion configurations — half-integer relational closure — interact with the projection geometry in a context where a time direction exists. This is a candidate direction, not a derivation. It is consistent with the topos framework already in use for fractional charge. The formal account — showing that projection of half-integer closure configurations into a time-directed spacetime produces the correct chiral coupling — does not yet exist.

The Borde-Guth-Vilenkin Theorem. The BGV theorem proves that any spacetime with an average positive expansion rate must have had a past boundary. The Medium holds that expansion is a view-layer artifact and therefore the theorem's premises do not hold at the substrate level. That response is coherent but incomplete. It does not show that the view layer fails to recover a structure to which BGV would apply. If projection recovers an expanding spacetime — as it must, since observers measure expansion — then the projected structure may satisfy BGV's premises even if the substrate does not. Whether the projected spacetime has a past boundary under BGV, and what that boundary means in substrate terms, is an open question.

The Cosmic Microwave Background and Big Bang Nucleosynthesis. The Medium offers an alternative account of cosmological redshift and rejects the interpretive framework of the Big Bang. That position requires alternative accounts of two independent observational pillars — and neither is yet given. The CMB presents three constraints: a near-perfect blackbody spectrum at 2.725 K, requiring a hot opaque early phase; uniformity across causally disconnected regions; and anisotropies that match the seeding of large-scale structure. Big Bang nucleosynthesis presents a fourth: the observed abundances of hydrogen, helium, deuterium, and lithium match BBN predictions with extraordinary precision, requiring specific early temperature and density conditions. These are independent of the three view-layer assumptions The Medium addresses. The Medium's redshift account addresses none of them. These are the most consequential open problems in the cosmological section. Both are carried explicitly here and deferred to future work.

Color Charge. The strong force operates via color charge — red, green, blue, and their anticolors. Three values. The SU(3) group structure of the strong force is built on this three-fold symmetry. The Medium's substrate is binary: ś1. Two values. Nothing in the current framework obviously produces a three-valued structure. Two candidate paths. First: color charge is not a third substrate value but a relational property of configuration geometry. A single quark has no independent color — color is defined by how three quarks relate to each other within a baryon. Color neutrality is the only physical reality; individual quark colors are gauge labels with no substrate reality. The three-fold structure is a labeling convention for a three-quark relational closure, not a third substrate character. This connects directly to the gauge symmetry entry: SU(3) color is the symmetry group of the freedom to relabel color assignments without changing physics. If the substrate's co-manifestation web has no preferred global labeling, SU(3) color may emerge as one expression of that freedom at the view layer. Second: color charge requires a substrate extension — a third distinction value or a two-level character structure — that the current framework does not have. Option one is more consistent with the existing framework and is the candidate direction. The formal account does not exist.

A stronger possibility: color is not needed at the view layer either. Color is never directly observed — confinement guarantees that no colored object reaches a detector. Every observed state is color-neutral. Color was introduced to explain why only certain combinations are stable, without access to substrate structure. If the relational web already accounts for why three-quark and quark-antiquark closures are the stable forms, color does no additional work at the view layer. The SU(3) gauge freedom confirms this: the freedom to relabel red/green/blue without changing any observable is the mathematical statement that color is not a physical property. It is a labeling convention. Conventions can be dissolved. The current treatment of color as unassigned assumes it must be placed somewhere in the framework. The stronger question is whether it needs to be placed at all, or whether it dissolves as a Standard Model artifact of describing hadronic stability from collider observations rather than from substrate structure. This should be evaluated before attempting derivation.

Dark Matter. Dark matter interacts gravitationally but not electromagnetically, not via the strong force, and at most weakly. It is invisible, stable, and accounts for roughly five times more mass than visible matter. In The Medium's terms, a dark matter candidate is a stable Planck bit configuration at a $\mathcal{K}$ extremum with nonzero N (mass — gravitational coupling at the view layer), zero net distinction character (electrically neutral), and no relational closure that engages the projection channels producing electromagnetic or strong force coupling. It is a pattern that achieves stability under the adjacency constraint but does not light up the observer's EM or strong projection channels. From the substrate perspective, dark matter is not mysterious — it is a class of stable configurations that happen not to couple to the forces visible matter couples to. The open derivation: identify which specific Planck bit patterns satisfy the adjacency constraint, achieve a $\mathcal{K}$ extremum, carry zero net distinction character, and have no EM or strong relational closure. Those patterns are the dark matter candidates. Whether the framework produces candidates matching the observed mass distribution and halo structure is the empirical test.

Color Charge, Gravity, and the Unassigned Category. Color charge, gravity, neutrino oscillation, and dark matter have not been placed with confidence in either layer. Gravity is the most consequential.

General relativity identifies gravity with spacetime curvature. The Medium makes spacetime view-layer emergent. If spacetime is view-layer, gravity as spacetime curvature is also view-layer. That placement has three consequences worth naming.

First, a potential strength: the equivalence principle — inertial mass equals gravitational mass to extraordinary precision — may be automatic. If mass corresponds to N and N governs both inertial resistance and gravitational coupling, the equivalence principle is a structural consequence of one substrate quantity projecting two ways. It stops being a mysterious coincidence.

Second, an open problem: quantum gravity. Every other force has been quantized. Gravity resists. If all four forces are view-layer, quantum gravity remains a view-layer problem and The Medium does not obviously resolve it. The possible path: gravity may not be like the other forces. It may bridge both layers — substrate configuration density shaping projection geometry, projection geometry feeding back on substrate structure. That bridge is the candidate direction. It is not developed.

Third, gravitational waves propagate through spacetime. If spacetime is view-layer, gravitational waves are view-layer phenomena. LIGO detects them. That is consistent with the placement.

Whether color charge is substrate or view-layer emergent; how dark matter candidates arise from substrate structure — these remain at the frontier. Dark energy is not a frontier question. It dissolves when redshift is correctly understood.

A candidate direction for gravity emerged from the iota packing model (developed 2026-05-23). Space is relational — exists as relationships between ι events, not as a void or independent fabric. Spacetime curvature is a large-scale pattern of ι alignment. A gravity well is a persistent local deviation in packing that the surrounding medium tries to correct.

Mass is the stable stochastic imperfection. Gravity is the strain field. The ι packing model gives infinite reach without a carrier particle: every ι event participates in the ground state, so every stable imperfection propagates strain through the medium without limit and without signal. The equivalence principle is automatic — the same N that resists inertial change is the N whose packing imperfection generates the strain field.

Quantum gravity's resistance to derivation is consistent with this account. Gravity is not like the other forces at the view layer because it is not a view-layer force in the same sense. It bridges substrate packing geometry and view-layer spacetime. Attempting to quantize it applies view-layer tools to something that only exists because view-layer granularity was averaged away. The failure is structural, not mathematical.

This direction requires: showing how the strain field through the ι medium maps onto the inverse square law; determining what constrains the iota's relational mode count; and connecting the packing geometry to the projection mechanism that produces the spacetime manifold. None of these derivations exist. The direction is named.

The Observer Boundary. The observer is a sustained binary configuration within ϕ, attempting to describe the substrate that produced it from within. Every formal system contains truths it cannot prove from within itself. The Medium faces an analogous boundary and acknowledges it.

9 Summary

10 A Personal Note

I am seventy-nine years old. I am not an academic. I have no credentials in physics or mathematics. I began this work with a question that would not leave me alone and a conviction that the question deserved a serious attempt at an answer.

I began this knowing I may not live to see it reach its conclusions. That does not trouble me. What matters is that the ideas are documented, attributed, and made available to anyone who finds them worthy of development. If this framework contains something true it belongs to whoever can take it further. My name attached to it is enough — for the record, and for my family.

Gregor Mendel published his work on inheritance and it sat unread for decades before being recognized as foundational. The ideas belong to whoever finds them. The record belongs to their origin.

11 License and Attribution

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Table 1: Complete symbol table for version 0.5

Symbol	Definition
ι	The Medium's act of self-distinction — produces character as the difference the dist
$\phi : M \to \{+1, -1\}$	Distinction character field — records characters produced by ι events
$x \sim y$	Co-manifestation — shared state, no spatial relationship
$\kappa = \sum \iota$	Coupling character — net distinction character of a local cluster
$\mathcal{K} = \sum \kappa$	Configuration energy — governs stability via extremum condition
$\hat{H} = \text{proj}(\mathcal{K})$	Hamiltonian — projection of $\mathcal{K}$ into spacetime
N	Count of ι events in a configuration — first approximation for mass
$\text{sgn}(\mathcal{K})$	Net distinction character of configuration energy — projects to charge
Spin	Relational closure of encounter pattern within the web
Adjacency constraint	Identical adjacent distinction characters forbidden — source of minimum threshold
Projection boundary	Planck scale — where dimensionless substrate meets dimensional observer
$\iota_{\text{proj}} = 1$	One projected ι event — the Medium's unit of action
c	Projection-layer ratio of projected distance to projected time
G	Gravitational constant — substrate derivation not yet complete
$x \sim y \to$ entanglement	Co-manifestation projects to quantum entanglement
Entropy	Projection-layer emergent — observer's coarse-graining of microstates
Time	Projection-layer emergent — ordered account of substrate transitions
Redshift	Three-layer projection artifact — not recession velocity
Color charge	May dissolve as labeling artifact; never directly observed (confinement)
Dark matter	Stable config: nonzero N, zero net character, no EM/strong closure
Dark energy	Dissolves as projection artifact of misread redshift